1. Title of the Project / Product

‘SwasthEdge’ A Low-Cost Self-Diagnosing Edge Controller for Industrial Sensor Reliability

2. Problem Statement

Low-cost edge controllers in industrial and remote environments can face sensor drift, disconnection, abnormal power consumption, and supply instability, creating a need for an embedded system that can monitor device and sensor health, distinguish persistent faults from temporary abnormalities, and provide an appropriate local response.

3. Solution Proposed

‘SwasthEdge’ is a low-cost self-diagnosing edge controller that monitors both connected sensors and controller health to detect faults early and respond locally.

Key Features:

- **Sensor-drift detection** using lightweight statistical methods such as EWMA/CUSUM.
- **Power-health monitoring** using voltage and current sensing.
- **MCU health monitoring** using temperature, reset, watchdog, communication, and memory checks.
- **Health-fusion logic** combining sensor and device-health data to generate a **fault-confidence value**.
- **Graceful degradation** with Normal, Warning, Degraded, and Safe operating states.
- **Local diagnostic logging** to store fault information when network connectivity is unavailable.

Technology: Low-cost MCU such as **STM32**, INA219/INA226 for current monitoring, an industrial-representative sensor, and local storage.

4. Implementation Plan

- **Target Beneficiaries**: Small and medium-scale industries, manufacturing systems, smart agriculture and environmental monitoring, remote infrastructure, and industrial IoT/edge deployments where continuous network connectivity or manual monitoring may not always be available.
- **Deployment Approach**: ‘SwasthEdge’ will be implemented using a low-cost MCU, representative industrial sensor, and voltage/current monitoring circuitry. Normal sensor and controller behaviour will first be established, after which the system will continuously monitor sensor and device health and detect abnormal conditions locally.

- **Expected Outcomes**: The system will demonstrate early detection of sensor and controller faults, distinguish persistent faults from temporary abnormalities, provide appropriate local responses through Normal, Warning, Degraded, and Safe states, and maintain diagnostic information even when network connectivity is unavailable.
- **Deliverables**: A working MCU-based 'SwasthEdge' prototype with embedded self-diagnosis firmware; a controlled fault-injection setup covering sensor drift, disconnection, power disturbance, abnormal current consumption, and communication failure; health-fusion and state-management logic; local diagnostic logging; and a test report covering detection performance and embedded resource usage.

5. Safety and Efficiency Analysis

- **Safety:** 'SwasthEdge' reduces the risk of undetected sensor and controller faults by verifying abnormal behaviour using sensor readings along with voltage, current, and internal health indicators before increasing fault confidence. Depending on the confirmed condition, the system can issue a warning, isolate a faulty sensor, use a fallback value, reduce operation, attempt communication recovery, or enter a predefined safe state.
- **Efficiency:** Lightweight statistical methods will be evaluated before more computationally expensive models. Flash usage, RAM usage, CPU utilization, processing latency, and additional power consumption will be measured to ensure the self-diagnosis functions remain suitable for low-cost embedded systems.

6. Scalability and Future Development

- **Scaling Plan**: SwasthEdge will begin with a single prototype for controlled validation and can later be extended to multiple sensor nodes, different wired/wireless interfaces such as RS485, LoRa, Wi-Fi, and BLE, and a common diagnostic-event format for integration with larger monitoring systems.

- **Future Enhancements**: Adaptive baselines for different operating conditions, TinyML-based anomaly detection, additional fault classes such as memory and communication faults, advanced sensor fusion, multi-node health monitoring, fleet-level analysis, and a compact custom PCB implementation

7. Innovation Advantage

'SwasthEdge' combines sensor-drift evidence with electrical and controller-health data in a low-cost edge controller to confirm faults locally and support graceful degradation.

Key Differentiators:

- Local diagnosis without continuous cloud connectivity.
- Combines sensor, voltage/current, and controller-health information.
- Detects early degradation as well as complete sensor failure.
- Uses lightweight methods suitable for resource-constrained MCUs.
- Generates a fault-confidence value instead of relying only on a single threshold.
- Supports graceful degradation and predefined local responses.
- Stores diagnostic evidence locally.
- Measures computational and power overhead.
- Validates faults using controlled physical fault injection.

Research Gap: Existing approaches often study sensor faults, anomaly detection, or power monitoring separately. 'SwasthEdge' combines these within a low-cost MCU-based diagnostic layer, with emphasis on fault confirmation, false alarms, resource overhead, and local response.

Validation: The project will compare sensor-only detection with multi-parameter health fusion to evaluate whether combining independent health indicators improves fault confirmation and reduces inappropriate responses.

8. Visual Impact & User Journey

- **User journey:** Existing edge system → quick retrofit → continuous health monitoring → early fault detection → local response → graceful operation → reduced downtime and maintenance risk.

